



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to D.N.V. FUEL STATION

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

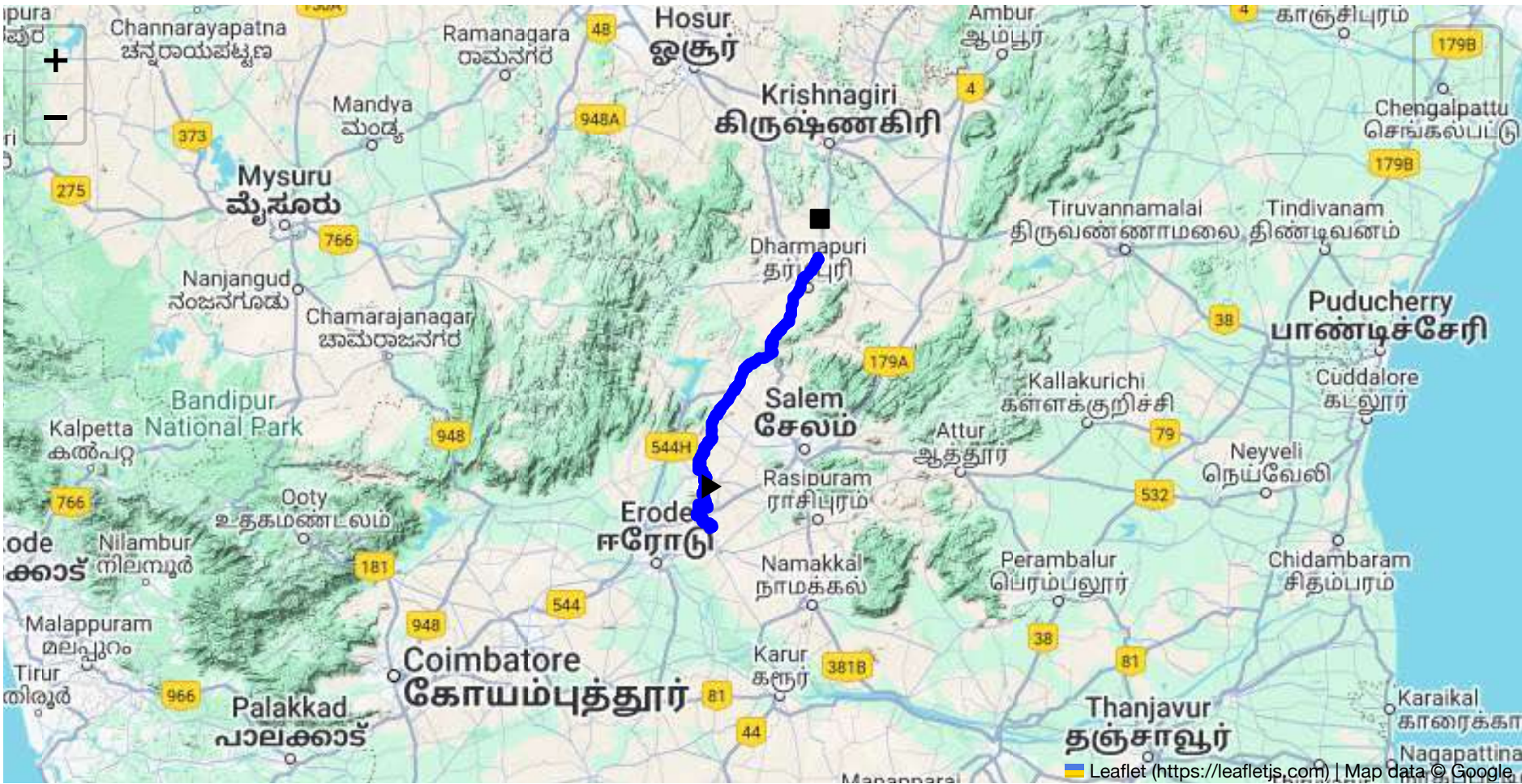
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 111.93 km
Estimated Duration: 2.6 hours
Adjusted Duration (Heavy Vehicle): 3.2 hours

Start: (11.4381, 77.8734)
End: (12.1976273, 78.18177701)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The journey from CVQF+23W, Sangagiri to 55WJ+XJG, Dharmapuri, spans approximately 111.93 kilometers. The route passes through several towns including Sankari West and Jaijainagra, primarily traversing state highways and some rural roads. The terrain is relatively flat, but there are areas of hills and forest, particularly as you approach Dharmapuri.

2. Typical Weather Conditions and Potential Weather-Related Hazards

- Weather Conditions:** The region experiences a tropical climate with hot, dry summers and mild winters. The monsoon season, from June to September, can bring heavy rains.
- Weather-Related Hazards:** The monsoon season can lead to waterlogging and reduced road visibility. Fog might be an issue during the early mornings, especially in the hilly sections near Dharmapuri.

3. Analysis of Traffic Patterns

- Peak Hours:** Traffic congestion is more pronounced between 8 AM to 10 AM and 5 PM to 8 PM, particularly near towns like Sangagiri and Sankari West.
- Congestion-Prone Areas:** Expect delays near market areas and junctions in small towns where roads are narrower.

4. Assessment of Road Quality and Infrastructure

- Road Condition:** Roads are generally well-maintained but can have sporadic potholes and narrow sections, particularly on rural bypasses.

- **Infrastructure:** Some sections may lack side barriers and have limited street lighting. The main routes are paved, but auxiliary roads might be under construction or damaged post-monsoon.

5. Suggestions for Alternative Routes for Emergencies

During emergencies, alternative routes include:

- Using NH44 which is more direct and often in better condition, although farther from the actual route.
- Redirecting via smaller state roads away from towns to avoid congested centers.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Transporting hazardous materials in Tamil Nadu requires compliance with specific safety regulations, including vehicle permits, scheduled transport times to minimize public exposure, and ensuring drivers have adequate training in handling such materials.

7. Overview of Historical Incidents

There have been occasional incidents involving heavy vehicles, mainly due to mechanical failures or weather conditions. The most common issues are road spills during monsoon rains.

8. Environmental Considerations and Sensitive Areas

- **Sensitive Areas:** Several villages and farmland areas along the route require cautious driving to prevent accidents and environmental damage.
- **Environmental Considerations:** Ensure proper disposal of waste and materials to avoid spills in farming areas.

9. Analysis of Communication Coverage

- **Coverage:** Most of the route is covered by major telecom providers, but there are potential dead zones in rural and hilly sections, especially towards Dharmapuri.
- **Suggestions:** Equip vehicles with multiple communication devices and keep emergency contact numbers accessible.

10. Estimated Emergency Response Times

- **Sangagiri to Sankari West:** Approximately 30-40 minutes due to closer vicinity to larger towns.
- **Sankari West to Dharmapuri:** Can take up to 60 minutes, factoring in poorer road conditions and rural sections.

11. Overall Summary of Risk Assessment

The route from Sangagiri to Dharmapuri presents several risks, notably during the monsoon due to wet roads and reduced visibility. Traffic congestion and road quality vary and can impact heavy vehicles, particularly those carrying hazardous materials. Emergency preparedness involves planning for communication coverage gaps and ensuring compliance with local regulations. However, with proper planning and adherence to safety protocols, these risks can be effectively managed.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.58300, 77.85813	15 KM/Hr
1	Roundabout	High	11.60348, 77.83967	15 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	High	11.44900, 77.87406	15 KM/Hr
5	Turn	High	11.45353, 77.85697	15 KM/Hr
6	Turn	Medium	11.45781, 77.85203	30 KM/Hr
7	Turn	High	11.46307, 77.84941	15 KM/Hr
8	Turn	High	11.46309, 77.84683	15 KM/Hr
9	Turn	High	11.46362, 77.84681	15 KM/Hr
10	Turn	Medium	11.46404, 77.84728	30 KM/Hr
11	Turn	Medium	11.46506, 77.84764	30 KM/Hr
12	Turn	Medium	11.46708, 77.84710	30 KM/Hr
13	Turn	Medium	11.46919, 77.84499	30 KM/Hr
14	Turn	High	11.47052, 77.84515	15 KM/Hr
15	Turn	Medium	11.47333, 77.83976	30 KM/Hr
16	Turn	Medium	11.47319, 77.83889	30 KM/Hr
17	Turn	High	11.47599, 77.83429	15 KM/Hr
18	Turn	High	11.48030, 77.83486	15 KM/Hr
19	Turn	Medium	11.48074, 77.83599	30 KM/Hr
20	Turn	Medium	11.49399, 77.83833	30 KM/Hr
21	Turn	High	11.49482, 77.83896	15 KM/Hr
22	Turn	High	11.50415, 77.83805	15 KM/Hr
23	Turn	High	11.50438, 77.83843	15 KM/Hr
24	Turn	High	11.49948, 77.84585	15 KM/Hr
25	Blind Spot	Blind Spot	11.49859, 77.86557	10 KM/Hr
26	Turn	Medium	11.50909, 77.86025	30 KM/Hr
27	Turn	Medium	11.52561, 77.85522	30 KM/Hr
28	Turn	Medium	11.52766, 77.85585	30 KM/Hr
29	Turn	Medium	11.54721, 77.85263	30 KM/Hr
30	Turn	Medium	11.55065, 77.85515	30 KM/Hr
31	Turn	Medium	11.55636, 77.85791	30 KM/Hr
32	Turn	Medium	11.56395, 77.85969	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
33	Turn	High	11.57903, 77.85432	15 KM/Hr
34	Turn	Medium	11.58092, 77.85775	30 KM/Hr
35	Turn	Medium	11.58672, 77.85835	30 KM/Hr
36	Turn	Medium	11.60943, 77.84147	30 KM/Hr
37	Turn	Medium	11.63116, 77.84077	30 KM/Hr
38	Turn	Medium	11.64036, 77.84589	30 KM/Hr
39	Turn	Medium	11.64155, 77.84858	30 KM/Hr
40	Turn	Medium	11.64570, 77.85137	30 KM/Hr
41	Turn	Medium	11.67280, 77.86013	30 KM/Hr
42	Turn	Medium	11.67387, 77.86245	30 KM/Hr
43	Turn	High	11.68484, 77.86886	15 KM/Hr
44	Turn	High	11.68472, 77.87281	15 KM/Hr
45	Turn	High	11.69458, 77.87589	15 KM/Hr
46	Turn	Medium	11.69520, 77.87526	30 KM/Hr
47	Turn	High	11.69549, 77.87446	15 KM/Hr
48	Turn	High	11.76020, 77.89070	15 KM/Hr
49	Blind Spot	Blind Spot	11.75947, 77.89214	10 KM/Hr
50	Turn	Medium	11.83102, 77.94360	30 KM/Hr
51	Turn	Medium	11.83220, 77.94374	30 KM/Hr
52	Turn	Medium	11.89740, 77.98073	30 KM/Hr
53	Turn	Medium	11.91914, 78.01331	30 KM/Hr
54	Turn	Medium	11.91866, 78.02552	30 KM/Hr
55	Turn	High	11.93616, 78.05389	15 KM/Hr
56	Turn	High	11.93704, 78.05402	15 KM/Hr
57	Turn	Medium	11.95594, 78.04730	30 KM/Hr
58	Turn	Medium	11.95818, 78.04978	30 KM/Hr
59	Turn	Medium	11.95876, 78.04990	30 KM/Hr
60	Turn	High	11.96035, 78.04946	15 KM/Hr
61	Turn	Medium	11.96304, 78.05136	30 KM/Hr
62	Turn	Medium	11.96399, 78.05135	30 KM/Hr
63	Turn	Medium	11.96458, 78.05084	30 KM/Hr
64	Turn	Medium	11.96496, 78.04988	30 KM/Hr
65	Turn	High	11.96580, 78.04923	15 KM/Hr
66	Turn	Medium	11.96675, 78.04939	30 KM/Hr
67	Turn	Medium	11.96791, 78.05070	30 KM/Hr
68	Blind Spot	Blind Spot	12.20336, 78.18465	10 KM/Hr
69	Turn	High	12.19810, 78.18155	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Parusuraman Hospital	11.697164, 77.877792	30 km/h	Medium
1	hospital	JP Nature Cure Hospital Siddha and Naturopathy	11.7002949, 77.8770897	30 km/h	Medium
2	hospital	SSS Hospital	11.7501471, 77.8832809	30 km/h	Medium
3	clinic	DSR Clinic	11.7603176, 77.8908853	30 km/h	Medium
4	hospital	Srilakshmi Hospital	11.7601, 77.892272	30 km/h	Medium
5	hospital	Government Hospital	11.758158, 77.894528	30 km/h	Medium
7	hospital	Vasu Speciality Hospital	11.8331097, 77.943169	30 km/h	Medium
8	hospital	Rajamanickam Hospital	11.8332408, 77.9431081	30 km/h	Medium
9	hospital	Child Hospital	11.833001, 77.943415	30 km/h	Medium
10	hospital	Naveen Nursing Home	11.8335355, 77.9437441	30 km/h	Medium
11	hospital	Selvakumar Hospital	11.836215, 77.945842	30 km/h	Medium
12	clinic	SK Clinic	11.83619, 77.9457752	30 km/h	Medium
13	hospital	Sri Sakthi Hospital	11.835714, 77.9451014	30 km/h	Medium
14	clinic	Sri Krishna Clinic	11.835443, 77.9450587	30 km/h	Medium
15	hospital	Thoppur, Government Hospital	11.9398824, 78.0551658	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
6	college	Kavery Educational Institutions	11.8255319, 77.9342256	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
16	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.58300, 77.85813



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.60348, 77.83967



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45781, 77.85203



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46309, 77.84683



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46362, 77.84681



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46404, 77.84728



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46506, 77.84764



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46708, 77.84710



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46919, 77.84499



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47052, 77.84515



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.47333, 77.83976



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.47319, 77.83889



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47599, 77.83429



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.48030, 77.83486



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.48074, 77.83599



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49399, 77.83833



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49482, 77.83896



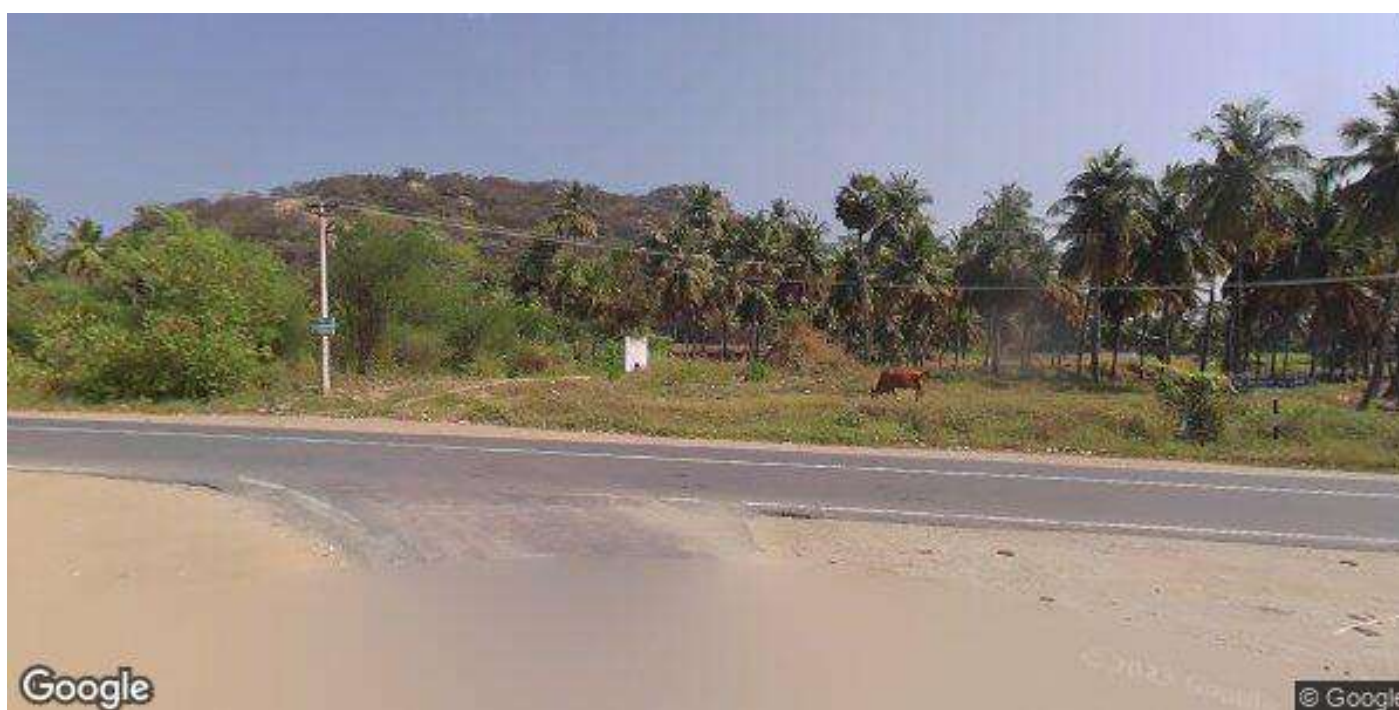
Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.50415, 77.83805



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.50438, 77.83843



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49948, 77.84585



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49859, 77.86557



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50909, 77.86025



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52561, 77.85522



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52766, 77.85585



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.54721, 77.85263



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55065, 77.85515



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55636, 77.85791



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56395, 77.85969



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.57903, 77.85432



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.58092, 77.85775



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.58672, 77.85835



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60943, 77.84147



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63116, 77.84077



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64036, 77.84589



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64155, 77.84858



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64570, 77.85137



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67280, 77.86013



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67387, 77.86245



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.68484, 77.86886



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.68472, 77.87281



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69458, 77.87589



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69520, 77.87526



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69549, 77.87446



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.76020, 77.89070



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.75947, 77.89214



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.83102, 77.94360



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.83220, 77.94374



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.89740, 77.98073



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91914, 78.01331



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91866, 78.02552



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.93616, 78.05389



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.93704, 78.05402



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95594, 78.04730



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95818, 78.04978



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95876, 78.04990



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96035, 78.04946



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96304, 78.05136



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96399, 78.05135



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96458, 78.05084



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96496, 78.04988



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96580, 78.04923



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96675, 78.04939



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96791, 78.05070



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.19810, 78.18155
